EVCSec: Adaptive Authentication Approach for Electric Vehicle Charging System*

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Abstract

The rapid expansion of electric vehicle charging infrastructure, combined with equipment heterogeneity from multiple manufacturers, diverse device environments, and public network reliance, has escalated security threats against EVCS. Legacy protocols such as OCPP 1.6 and ISO 15118 expose vulnerabilities in static authentication, session management, and certificate handling, with real-world incidents including OCPP zero-day flaws and global EV charging network breaches. This paper proposes an efficient, adaptive reputation-based authentication protocol for EVCS environments. Built on an integrated ISO 15118-20 and OCPP 2.0.1 architecture, it computes reputation scores from vehicle charging history, anomaly frequency, time-based trust decay, and location risk, classifying them into three levels for dynamic authentication strength adjustment. Authentication levels adapt across EV-charger-CSMS segments, ensuring real-time session continuity and data integrity. Formal verification via Scyther confirms security against known attacks.

1 Introduction

The rapid proliferation of electric vehicles (EV) is accelerating the transition to sustainable transportation systems, with 14 million new EVs registered worldwide in 2023, reaching a total of 40 million units. This growth in the EV market necessitates the expansion of Electric Vehicle Charging

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Systems (EVCS), which have become a core component of modern smart transportation infrastructure [1]. EVCS not only provides physical power supply to electric vehicles but also delivers intelligent services such as remote monitoring, charging session management, and billing based on Internet of Things (IoT) technology, functioning as a Cyber-Physical System (CPS). EVCS forms a complex ecosystem consisting of Charge Points (CP), Central System Management Systems (CSMS), power grids, and user electric vehicles, with interactions between these components conducted through standardized communication protocols [2]. Open Charge Point Protocol (OCPP) is the most widely used open communication standard protocol in EV charging infrastructure, developed by the Open Charge Alliance (OCA). OCPP standardizes communication between charge points and central management systems to ensure interoperability and provides core functions such as remote monitoring, charging session management, firmware updates, and smart charging [3-4].

Currently, OCPP 1.6 and OCPP 2.0.1 versions are mainly used, with OCPP 1.6 implemented in JSON over WebSocket format to support lightweight message transmission. OCPP 2.0.1 introduces enhanced security features to address security vulnerabilities of previous versions, including X.509 digital certificate support for mutual authentication, certificate management messages, and firmware update security through electronic signatures. As IoT devices connected to the internet, EVCS are exposed to various cyber security threats, which can seriously impact not only individual charging stations but also the stability of the entire power grid and user personal information protection. Recent studies report various security threats that may occur in EVCS environments [5].

Recent EVCS security incidents are materializing as real threats. In February 2024, the UK government suspended sales of Spanish-made EV chargers due to cybersecurity concerns, indicating potential threats that charging infrastructure can pose to national energy security [6]. In April 2024, six zero-day vulnerabilities (CVE-2024-23971, etc.) were discovered in the OCPP protocol, exposing security weaknesses in backend communication of numerous charging management systems [7]. Particularly, ChargePoint Home Flex chargers were found to have remote code execution vulnerabilities due to lack of user input string validation in OCPP message processing. In August 2024, the U.S. government launched a cybersecurity initiative for clean energy system protection, designating EV charging infrastructure and power grid security as top priorities [6]. Most seriously, in November 2024, a large-scale data breach occurred in a global EV charging network, compromising approximately 116,000 personal information, payment details, vehicle VINs, and charging station locations, affecting charging operators in multiple countries [3]. These security vulnerabilities seriously undermine the trustworthiness and safety of EVCS, posing direct threats to power grid stability and user personal information protection. OCPP vulnerabilities mainly stem from weak authentication mechanisms, lack of encryption, and inadequate certificate verification, demonstrating the limitations of static authentication methods. Therefore, developing dynamic and adaptive authentication mechanisms to strengthen EVCS security is urgently needed. The EVCS environment targeted in this paper is configured as shown in Figure 1 and has characteristics that make it difficult to provide sufficient security with existing static authentication methods alone [5, 8]:

(1) Heterogeneous environment with mixed chargers from various manufacturers and models
(2) Characteristics exposed to physical attacks due to installation in public places
(3) Requirements to maintain persistent connections between charging stations and central systems
(4) Low-latency requirements for real-time charging service provision

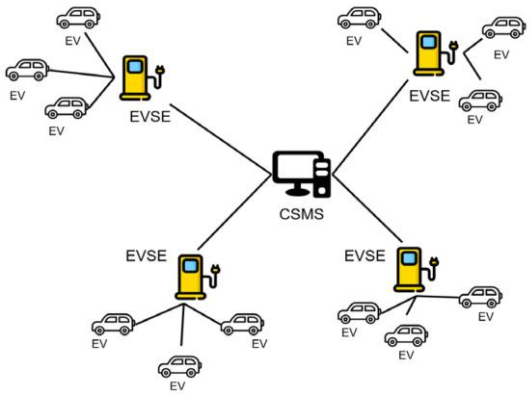


Figure 1: EVCS Environment Architecture

ISO 15118 defines communication protocols between electric vehicles and chargers, including Plug & Charge, bidirectional energy transfer, and enhanced cyber security systems. ISO 15118 provides robust security features such as mutual authentication through X.509-based PKI (Public Key Infrastructure) between EVCC (Electric Vehicle Communication Controller) on the EV side and SECC (Supply Equipment Communication Controller) on the charging station side, TLS-based encrypted communication, session key exchange, and certificate lifecycle management. This security framework effectively prevents unauthorized access, data tampering, replay attacks, etc., between EVs and charging stations and protects user privacy [8].

This study aims to propose a security protocol that simultaneously improves the security and practicality of EVCS by integrating adaptive authentication techniques into the OCPP protocol and ISO 15118 protocol. The main contributions of this paper are as follows:

(1) Proposes a reputation-based adaptive authentication protocol for EVCS environments, which comprehensively considers charging station's past behavior history and current communication context to dynamically adjust authentication levels.

(2) The proposed protocol is designed to be compatible with existing OCPP message structures, enabling gradual deployment to existing EVCS infrastructure.

(3) Uses the formal verification tool Scyther to formally verify the security properties of the proposed adaptive authentication protocol and prove security against various attack scenarios.

(4) Confirms through performance evaluation that the proposed technique provides practical performance along with enhanced security.

The organization of this paper is as follows. Section 2 analyzes OCPP protocols widely used in EVCS communication and existing studies for improving EVCS security. Section 3 designs the reputation-based adaptive authentication protocol, Section 4 presents security evaluation and performance analysis results of the adaptive authentication protocol proposed in Section 3. Finally, Section 5 summarizes the research results and presents future research directions.

2 Related Work

2.1 OCPP

OCPP is based on a client-server model, where the CP serves as the client and the SMS serves as the server. The charge point is a physical electric vehicle charging device that includes one or more

connectors and provides data such as charging session information, connector status, and energy consumption. The CSMS communicates with charge points through OCPP to perform charging session monitoring, energy consumption data collection, and billing processing.

The OCPP communication flow is configured as follows [10]:

- Boot notification: When a charge point starts or restarts, it sends a registration request to the CSMS, providing identification information and configuration information.
- Status notification: The CSMS sends a status request to the charge point to collect real-time information such as availability, charging power, and ongoing charging sessions.
- Charging session management: The CSMS sends a start command to the charge point to initiate a charging session. The charge point acknowledges the command, starts the session, and sends status updates to the CSMS. The CSMS can send commands to pause, restart, or terminate the charging session.
- Transaction data transmission: The charge point sends transaction data such as start time, end time, energy consumption, and cost to the CSMS during and after the charging session.
- Firmware updates: The CSMS sends a firmware update request to the charge point, which performs the firmware update process and sends progress status and completion notifications to the CSMS.

OCPP consists of several functional blocks, each containing specific purposes and requirements:

- Security: Defines protocol security specifications, including authentication, encryption, certificate management, and security logging.
- Provisioning: Provides functionality for registering charge points to the network and retrieving basic configuration information.
- Authorization: Handles user authentication before charging session initiation, including authentication message processing and authentication caching.
- Local authorization list management: Provides functionality for managing local whitelists on charge points, enabling user authentication even in offline states.
- Transactions: Defines basic functions related to transactions started/stopped at charge points.
- Remote control: Includes functions such as remote transaction control, connector unlock, and remote trigger from the CSMS.
- Availability: Defines status notification message transmission functionality.
- Smart charging: Supports smart charging functionality using charging schedules.

OCPP 1.6 basically provides only transport layer security and HTTP basic authentication, and does not standardize certificate exchange or mutual authentication at the protocol level. OCPP 1.6 has several security vulnerabilities, with major issues as follows [3, 6, 11]:

- Weak authentication mechanism: OCPP 1.6 provides an option to allow authentication using only charge point identifiers, enabling attackers to easily impersonate legitimate charge points.
- Inadequate multi-connection handling: The OCPP standard does not clearly define how the CSMS should handle multiple WebSocket connections with the same charge point ID.

Consequently, attackers can intercept or continuously block legitimate charge point connections.

- Static authentication method: Existing OCPP uses static authentication methods that do not consider charge point's past behavior or current communication context, making adaptive responses to evolving threat environments difficult.

OCPP 2.0.1 overcomes these limitations by introducing strictly defined security profiles, key management for client-side certificates, secure firmware updates through signed firmware, and security event logs.

2.2 Literature Review

ISO 15118-20 [8] is the latest version defining 2nd generation network and application protocol requirements for the ISO 15118 standard. This standard includes functions such as wireless charging, bidirectional energy transfer (Vehicle-to-Grid, V2G), pantograph charging, etc., and uses enhanced security protocols of TLS 1.3 or higher. Smart charging and load management functions are highly advanced, and communication efficiency is significantly improved through dynamic certificate management and efficient message encoding. When an EV connects to a charger, a network and security connection is established between EVCC and SECC. Then mutual authentication is performed to verify trust on both sides, and charging service use and payment by the user are automatically authenticated and approved based on digital contract certificates embedded in the vehicle. Power specification exchange and periodic status reporting proceed after charging initiation, and finally the process is completed with charging termination request and session termination communication. All processes are securely handled through TLS encryption and integrity verification mechanisms. Meanwhile, it is not compatible with ISO 15118-2, requiring interoperability measures for different generations.

Hamdare et al. [13] proposed a comprehensive cyber defense framework to mitigate cybersecurity risks of OCPP. This study pointed out that OCPP, while being a widely adopted communication standard ensuring interoperability between EV charging stations and central management systems, is exposed to serious cyber risks due to weak authentication mechanisms and improper session handling. The framework proposed in provided targeted improvements to address OCPP security vulnerabilities and demonstrated performance surpassing existing OCPP versions and latest EV charging security solutions in terms of security and efficiency. This study also addressed difficulties in protocol adoption due to lack of backward compatibility with OCPP 2.0.1 and presented practical security enhancement measures.

P. Li et al. [14] propose a distributed defense model combining Zero Trust architecture and blockchain to address single points of failure and insider attack vulnerabilities of existing centralized trust models. This model applies Zero Trust principles to continuously verify all connections and requests, using ShangMi cryptographic algorithms to protect each stage of user registration, charging reservation, and payment. During user authentication, ring signature-based anonymous authentication protects personal information, and all transactions are recorded on blockchain to ensure tamper resistance and transparency. Certificates are managed through distributed PKI, charging reservation conditions are verified and automatic payment is executed through smart contracts, and network-wide transaction validity is verified through consensus mechanisms. This distributed architecture eliminates single points of failure and enhances overall system resilience and trustworthiness.

N. Hettiarachchi et al. [15] propose an anonymous authentication system based on 3GPP AKA protocol to address the problem that ISO 15118 Plug & Charge protocol exposes user ID and charging patterns, risking privacy breaches. This protocol uses AKA's one-way key functions (f1~f5) to generate MAC, expected response, anonymization key, encryption key, and integrity key required for mutual authentication, and anonymizes sequence numbers through XOR operations to prevent tracking. KEM

is used to securely exchange session keys between EV and eMSP, and AEAD encrypts authentication data and metering data to simultaneously ensure confidentiality and integrity. EV, eMSP, and charging stations each generate ECDSA signatures to verify message origin and integrity, and provide end-to-end security through multiple TLS connections and certificate chain verification. This design enables secure charging and payment without exposing user's actual identification information, achieving both privacy and security.

3 Adaptive Authentication Protocol for EVCS

In current electric vehicle charging infrastructure, EV and chargers EVCS perform mutual authentication through ISO 15118, while chargers and CSMS communicate via OCPP, forming a dual-layer structure. While this structure is secure in individual communication segments, the lack of linkage between the two protocols causes complexity in authentication management and session continuity issues. Additionally, each charging station and vehicle performing fixed levels of encryption creates inefficiency where authentication burden becomes excessive or unevenly distributed relative to attack difficulty.

Therefore, this paper proposes an adaptive authentication protocol that integrates the previously separated authentication structures of OCPP and ISO 15118-20 and applies adaptive mechanisms to adjust authentication levels according to the reputation of electric vehicles accessing charging stations.

The adaptive authentication protocol proposed in this paper does not perform verification independently when an EV sends an authentication request to EVCS. Instead, EVCS links with CSMS for real-time verification of certificate chains and reputation values. CSMS maintains reputation scores for each EV and dynamically evaluates trustworthiness by analyzing past connection history, charging patterns, location-based anomaly detection results, etc. Then, in the authentication procedure, lightweight encryption and single authentication are allowed for vehicles with high reputation, while enhanced authentication procedures are applied to terminals with low trustworthiness or attack potential. Through this, the system can dynamically adjust authentication strength to minimize overall process load while reducing attack surface and strengthening security.

As a result, the proposed adaptive authentication protocol combines ISO 15118-20's authentication framework, OCPP's security management functions, and reputation-based adaptive logic to satisfy scalability, reliability, and real-time requirements of EVCS operating environments.

3.1 Security Requirements

The security requirements of the adaptive authentication protocol proposed in this paper were derived based on ISO 15118-20, OCPP-related standards, and EVCS security research [8-15]. These summarize authentication, encryption, certificate management, privacy protection, and other requirements presented in prior research suitable for EVCS environments as follows:

(1) Mutual Authentication: Vehicles and charging stations must verify each other's legitimacy using trusted public PKI systems, with secure certificate exchange between EVCC and SECC. This protocol reflects this requirement by designing dynamic authentication mechanisms linked with reputation-based evaluation.

(2) Encrypted Communication: All messages must be protected from unauthenticated attackers. It must also prevent data eavesdropping, tampering, and replay attacks. Under adaptive authentication environments, additional encryption strength settings based on real-time risk analysis must be possible.

(3) Certificate Management: Full lifecycle management including certificate issuance, renewal, and revocation must be performed, complying with multi-stage PKI structures supporting Plug & Charge

authentication systems. This protocol can apply policies that strengthen certificate verification procedures for nodes with low reputation-based evaluation adaptively.

(4) Transaction Data Integrity Guarantee: Billing and charging session-related data can secure integrity through message signatures and HMAC verification. Additionally, data integrity verification levels can be dynamically adjusted in adaptive authentication to balance performance and security.

(5) Replay Attack Prevention: All communication sessions must include timestamps and nonces to prevent message reuse, and additional authentication procedures can be considered when abnormal patterns occur.

(6) Privacy Protection: During Plug & Charge processes, pseudonym-based authentication and anonymization techniques are applied instead of real names to minimize personal information exposure. Reputation-based and adaptive authentication techniques include methods that maintain trustworthiness while protecting user privacy.

(7) Reputation-based Adaptive Authentication: Analyzes past behavior history (reputation score) of charging stations and vehicles and current communication context (behavior patterns, location, network status, etc.) to dynamically adjust authentication procedure strength and steps. This enables real-time responses to evolving security threats while maximizing authentication process efficiency.

3.2 Reputation Score Model

This section presents the reputation score model, a core element of the reputation-based adaptive authentication protocol proposed for electric vehicle charging system environments. The reputation score quantitatively expresses each vehicle's trustworthiness by comprehensively considering past behavior, security event occurrence history, regional characteristics, time-elapsed trust decay, etc., and is used as a metric to dynamically adjust authentication strength (encryption level, number of authentication procedures, auxiliary verification criteria, etc.) when EVs access. Through this, both security and efficiency can be achieved simultaneously [15-16].

In this paper, the reputation score model is exemplified by expressing reputation scores in the 0~100 range, classifying vehicle trustworthiness levels into three grades as shown in Table 1. These reputation grades reference the digital fingerprint-based authentication threshold design in [17] and the three-level user risk classification method in [18]. Reputation grades used in Table 1 can be calculated through indicators and situation-based weights in Table 2.

Grade	Score	Type	Description
High	80~100	Lightweight Authentication	Trusted vehicles, safe areas, good history
Medium	50~79	Standard Authentication	General users, unverified areas
Low	0~49	Enhanced Authentication	Abnormal behavior or unauthorized attempts

Table 1: Reputation Score Model Grade (Example)

In the proposed reputation model, initial users or first-time EVs without prior charging history are handled conservatively. Since historical behavior cannot be used to establish trustworthiness, these users are assigned to the Low reputation grade by default and are required to perform enhanced authentication until sufficient normal charging records are accumulated.

Indicator	Definition	Value
R_history	Normal charging history	0~1
R_anomaly	Anomaly behavior detection history	0~1
R_time	Time-based trustworthiness	0~1
R_location	Location-based trustworthiness	0~1

Table 2: Reputation Score Model Indicator (Example)

To derive appropriate reputation scores according to situations, sufficient discussion and adjustment of trust grades, indicators, and weights are needed. Grades and categories used in grades can be derived through simulation and additional research, while indicators and weights can be initially set per charger and installation area but continuously monitored to analyze attack detection rates, false positive rates, etc., per charger or area for readjustment. Indicators in Table 2 can be defined as follows:

- Normal charging history (R_history): Represents the normal completion rate of charging sessions, assigning higher scores as session error or interruption rates decrease.
- Anomaly behavior detection history (R_anomaly): Evaluated based on network traffic anomaly detection, retransmission patterns, authentication failure events. Higher abnormal operation detection rates result in lower scores.
- Time-based trustworthiness (R_time): Trust gradually decreases based on elapsed time since the last normal charging.
- Location-based trustworthiness (R_location): Reflects charging station security certification status, regional risk levels, location history, etc. For example, public certified charging stations: 1.0, unauthorized private stations: 0.6, high-risk areas: 0.4.

Total weights are defined as $\sum_{i=0}^{n} w_i = 1$ to normalize indicator proportions. This paper uses 4 indicators total, with each weight modifiable by variables such as users, environment, etc. The weighted reputation score model is given by the following equation:

$$R = (w_1 * R_history\ + w_2 * R_anomaly\ + w_3 * R_time + w_4 * R_location)\ * 100$$

This paper presents three weight scenarios to express the impact of reputation score model weight changes on the protocol.

First, the uniform scenario assumes equal weights for all indicators, allowing equal reference to all indicators but potentially allowing malicious EVs to manipulate reputation or causing disadvantages to users affected by location/time decay.

Second, the behavior history scenario assigns high trustworthiness to users with long-term normal behavior history, reducing disadvantages from location/time decay in uniform scenarios, with increased anomaly detection weights enabling rapid grade adjustment upon abnormal behavior even with normal history.

Finally, the location scenario highly evaluates location-based indicators, allowing high reputation for safe infrastructure charging stations but potentially giving high trust grades to malicious EVs using safe areas or disadvantaging long-term normal users in risky areas. Table 3 shows weight examples applicable to these three scenarios.

Scenario	w_1	w_2	w_3	w_4
Uniform	0.25	0.25	0.25	0.25
Behavior History	0.35	0.3	0.15	0.2
Location	0.2	0.2	0.1	0.5

Table 3: Weight Adjustment by Scenario

Through the reputation score model presented in this section, each vehicle's trustworthiness level can be measured to adjust authentication levels and security mechanisms according to grades as in Table 4, expected to improve security and efficiency compared to existing methods [19-21].

Grade	Authentication Type	Security Mechanism
High	Lightweight Authentication	HMAC, etc.
Medium	Standard Authentication	ECDSA, etc.
Low	Enhanced Authentication	ZKP, Location verification, etc.

Table 4: Policy Changes by Grade

3.3 Protocol Design

This section proposes an adaptive authentication protocol that improves security and efficiency by integrating ISO 15118-20's Plug & Charge mechanism and OCPP 2.0.1's security management framework with the reputation score model. Major participants in the protocol are as follows:

- EV: Equipped with ISO 15118-20 based EVCC, possesses unique identifier ID_{EV}, contract certificate $Cert_{EV}$, private key SK_{EV}.
- EVCS: Performs ISO 15118-20 communication with EV through SECC, links with CSMS via OCPP 2.0.1, possesses unique identifier ID_{CP}, certificate $Cert_{CP}$, private key SK_{CP}.
- CSMS: OCPP backend server managing all charging stations and maintaining EV reputation database, possesses unique identifier ID_{CSMS}, certificate $Cert_{CSMS}$, private key SK_{CSMS}.
- Certificate Authority (CA): Issues and verifies certificates for all participants, manages PKI.

Figure 2 diagrams the proposed reputation-based adaptive authentication protocol, with detailed explanation as follows:

(1) EV and EVCS perform TLS 1.3 handshake to establish bidirectional secure communication channel. During this process, both entities exchange certificates, establish cryptographic keys for the session, ensuring confidentiality and integrity of transmission data.

(2) EVCS transmits AuthorizeReq message containing ID and certificate information provided by EV to CSMS. CSMS receives this request, queries EV's reputation score (REV), and determines EV's authentication level (Auth_Level) based on that score. If no prior reputation record exists for the EV, CSMS assigns the EV to the Low Level by default and applies enhanced authentication.

(3) CSMS transmits reputation score (REV), authentication level (Auth_Level), and challenge (Challenge) to be used in the authentication process to EVCS via AuthorizeResp message.

(4) EVCS relays this information back to EV so that EV can perform authentication procedures using the corresponding Challenge.

(5) EV can proceed with authentication according to the reputation score assigned to it. For high trust cases, lightweight authentication such as HMAC is performed; for medium trust levels, ECC-based ECDSA digital signatures can be used; for low trust cases, enhanced procedures including zero-knowledge proof (ZKP) authentication and additional verification such as location or biometric information can be performed. Mandatory policies according to reputation scores block attacks where malicious EVs disguise or select weak authentication methods.

(6) EVCS receives authentication response message from EV and transmits it back to CSMS to request final verification.

(7) CSMS reports authentication success or failure to EVCS based on this authentication result.

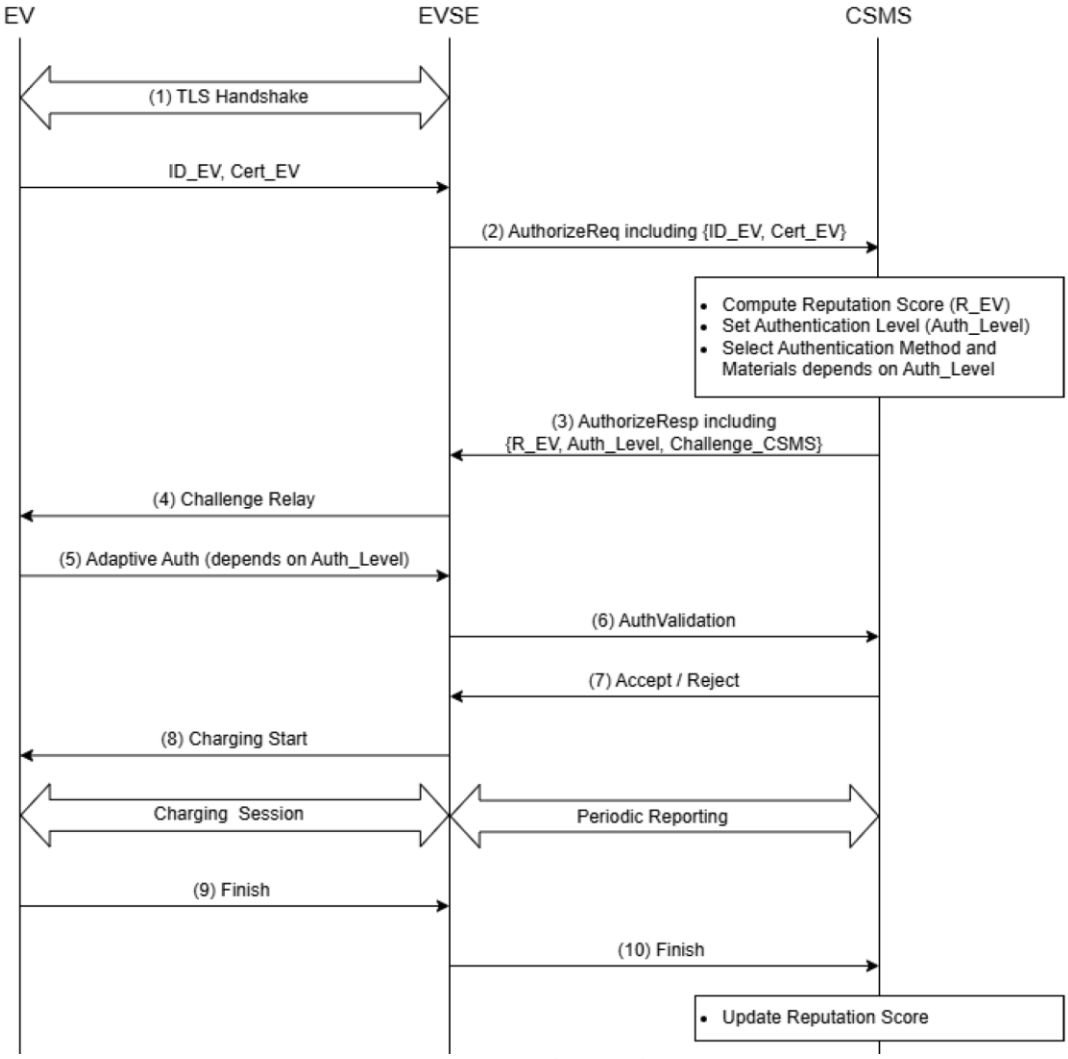


Figure 2: Proposed Protocol

Once authentication is successfully approved, (8) EVCS transmits charging start message to CSMS and initiates charging with EV. During charging, EVCS periodically reports meter values to manage charging status. Upon charging completion, (9-10) EVCS transmits final charging data along with termination event to CSMS. CSMS updates EV's reputation score based on this data for reflection in next authentication, and can adjust reputation scores to strengthen security if anomalies are detected.

4 Evaluation

This paper objectively evaluates the security and performance of the proposed reputation-based adaptive authentication protocol using the widely used formal verification tool Scyther [22]. In this verification process, the Dolev–Yao attacker model is adopted to analyze whether the proposed protocol

can withstand an adversary capable of intercepting, modifying, replaying, and injecting messages over a public communication channel. As specified in ISO/IEC 29128 [23-24], formal verification is recognized as an objective and reproducible method for evaluating security protocol trustworthiness. Additionally, the computational efficiency performance of the proposed authentication protocol is evaluated by comparing operational overhead with OCPP's basic authentication structure and EVCS security studies reviewed in Section 2.2.

4.1 Security Verification with Scyther

Scyther is an automated formal verification tool for security protocols developed by Cas Cremers, used to verify protocol security properties and detect potential attacks. The verification procedure through Scyther proceeds in 5 steps: (1) protocol modeling, (2) Claim Event definition, (3) model verification, (4) attack scenario identification, (5) results [25]. Since Scyther verifies the message flow and security properties of cryptographic protocols, the correctness of the reputation score calculation itself, the optimality of the threshold value, and the false positive rate are outside the scope of Scyther-based formal verification.

First, in the protocol modeling step, the cryptographic protocol to be analyzed is defined, including detailed information such as participants, roles, exchanged messages, and rules governing protocol behavior. It is written in SPDL (Security Protocol Description Language), a format language readable by Scyther.

In Claim Event definition, security properties of the protocol to verify are defined, which may include secrecy, authentication, integrity, etc. In model verification, automated analysis is performed on protocol participants and security behaviors. Claim Events defined in this paper are shown in Table 5.

Claim Event	Meaning
Alive	Guarantees that the entity completing communication has evidence that the other party executed the protocol at some past point.
Weakagree	Guarantees that the entity completing communication has evidence that the other party executed the protocol with it, but does not verify the other party's role or data exchanged.
Niagree	Guarantees that the entity completing communication has evidence that the other party executed the protocol with the correct role and agreed data set, but does not guarantee message order.
NIsynch	Guarantees that the entity completing communication has evidence that all messages were sent/received by the other party in the exact order specified in the protocol.

Table 5: Claim Event Deification

In scenario identification, it explores various possible scenarios and interactions between participants to check whether specified security properties are maintained or vulnerabilities exist. Finally, results generate a report containing information on vulnerabilities, security weaknesses, and identified attack scenarios.

Figure 3 shows the results of verifying the security of the adaptive authentication protocol proposed in this paper using Scyther. According to this, the adaptive authentication protocol proposed in this paper satisfies all Claim Events used in verification. Accordingly, the adaptive authentication protocol proposed in this paper is safe from attacks and proven to effectively protect charging infrastructure from various security threats when deployed in actual EVCS environments.

Claim				Status	Comments
adaptive_occp	EV	adaptive_occp,EV1	Alive	Ok	No attacks within bounds.
		adaptive_occp,EV2	Niagree	Ok	No attacks within bounds.
		adaptive_occp,EV3	Weakagree	Ok	No attacks within bounds.
		adaptive_occp,EV4	Nisynch	Ok	No attacks within bounds.
	EVSE	adaptive_occp,EVSE1	Alive	Ok	No attacks within bounds.
	CSMS	adaptive_occp,CSMS1	Alive	Ok	No attacks within bounds.
		adaptive_occp,CSMS2	Niagree	Ok	No attacks within bounds.
		adaptive_occp,CSMS3	Weakagree	Ok	No attacks within bounds.
		adaptive_occp,CSMS4	Nisynch	Ok	No attacks within bounds.

Figure 3: Scyther Verification Result

4.2 Performance Comparison

This section compares the proposed adaptive authentication protocol with existing studies analyzed in Section 2 in terms of operational overhead and qualitative aspects. The strength of the proposed adaptive authentication protocol lies in being able to dynamically adjust computational costs according to reputation scores. High-trust EVs receive High Level for high efficiency, while low-trust or first-time EVs receive Low Level for high security. This adaptive approach achieves optimal security-efficiency balance suited to situations, unlike existing protocols using fixed single security levels. Table 6 compares operational overhead by computation count with proposed protocol and existing studies [13-15].

Operation	[13]	[14]	[15]	Ours (H)	Ours (M)	Ours (L)
HASH	2	5	12	5	6	9
XOR	0	0	4	0	0	0
SYM	21	9	7	11	11	11
ASY	3	6	4	3	6	8
DS	0	6	6	2	2	2
CV	1	3	4	3	3	3

HASH: Hash operation; XOR: XOR operation; SYM: Symmetric key operation;
ASY: Public key operation; DS: Digital signature operation; CV: Certificate verification operation.
H: High Level (with HMAC option);
M: Medium Level (with ECDSA option);
L: Low Level (with ZKP option)

Table 6: Computational Overhead Comparison by Operation Count

The High Level of our proposed protocol incurs slightly higher cost than the lightest existing study but enhances security through strengthened certificate verification procedures. The Medium Level performs comparably to, while the Low Level provides the highest security among comparison targets. This demonstrates the flexible structure achieved through reputation-based adaptive approaches, delivering efficiency to high-trust EVs and enhanced security to low-trust EVs.

In Table 7, existing studies apply uniform security levels to all EVs using fixed methods, whereas this research achieves security-efficiency balance through reputation-based adaptive approaches. lacks

compatibility with existing OCPP infrastructure due to its blockchain-based separate architecture, and focuses solely on ISO 15118. This study integrates OCPP 2.0.1 and ISO 15118-20 to enhance practical deployment feasibility. Additionally, this research uniquely proves the security of the proposed authentication technique through formal verification using the well-known Scyther tool.

Aspect	[13]	[14]	[15]	Ours
Authentication Method	Fixed	Blockchain	Anonymous	Adaptive
Compatibility	△	×	△	○
Scalability	○	×	△	○
Privacy Protection	△	○	○	○
Real-time Performance	○	×	△	○
Formal Verification	×	×	×	○

Table 7: Qualitative Comparison

4.3 Attack Scenarios and Countermeasure

This section specifically analyzes how the proposed adaptive authentication protocol responds to major attack scenarios. Table 8 maps major attack types in EVCS environments with the proposed protocol's countermeasure mechanisms. Through analysis of the 7 major attack scenarios presented in Table 8, it is confirmed that the proposed adaptive authentication protocol comprehensively responds to major attack scenarios in EVCS environments through the combination of cryptographic foundations (TLS, PKI, digital signatures) and reputation-based dynamic mechanisms (adaptive authentication strength adjustment, anomaly detection). Particularly, its balanced design simultaneously achieves efficiency for normal users and enhanced security against malicious actors, expected to overcome limitations of existing simple static authentication methods.

Attack Scenario	Description	Countermeasure
Spoofing Attack	Malicious EV impersonates legitimate EV by stealing ID for unauthorized charging attempts	Detects sudden location changes and reclassifies reputation grade
Replay Attack	Attacker replays previously eavesdropped authentication messages to bypass authentication	CSMS verifies timestamp validity and nonce duplication upon message receipt
Man-in-the-Middle Attack	Attacker intercepts and tampers with communication between EV and EVCS	CSMS digital signature protects reputation score and authentication level
Denial of Service Attack	Attacker sends mass authentication requests to overload CSMS	Applies lightweight authentication to High grade EVs to reduce CSMS load and limits identical request frequency
Certificate Forgery Attack	Attacker creates fake certificates to impersonate legitimate EV	X.509 certificate chain verification
Reputation Manipulation Attack	Malicious EV mimics normal behavior to raise reputation score before attack	Real-time anomaly pattern detection using multiple indicators

Table 8: Major Attack Scenario and Countermeasure in EVCS Environment

5 Conclusion

Amid the rapid expansion of electric vehicle charging infrastructure and escalating security threats against EVCS, existing OCPP protocols revealed limitations in static authentication methods unable to adaptively respond to diverse threat environments. This study proposed a reputation-based adaptive authentication protocol integrating ISO 15118-20 and OCPP 2.0.1 standards with dynamic authentication mechanisms. The protocol computes reputation scores by synthesizing multiple indicators including vehicles' past charging history, anomaly detection frequency, time-based trust decay, and location-based risk assessment, dynamically adjusting authentication strength across three levels (High/Medium/Low). It ensures real-time session continuity and data integrity across EV-EVCS-CSMS segments while achieving optimal security-efficiency balance by applying lightweight authentication to high-trust vehicles and enhanced procedures to suspicious ones. Scyther formal verification confirmed satisfaction of all essential security claims (Alive, Weakagree, Niagree, Nisynch), proving resilience against major attacks including spoofing, replay, MITM, DoS, certificate forgery, and reputation manipulation. The protocol demonstrated computational efficiency comparable to or better than existing studies [13-15], with its adaptive structure providing practical superiority in real deployment scenarios.

Future research will integrate machine learning-based real-time anomaly detection, conduct statistical analysis of optimal reputation thresholds using large-scale EVCS data, expand support for bidirectional V2G scenarios, and develop standardized APIs for gradual migration from legacy OCPP 1.6 infrastructure, contributing significantly to building secure and efficient infrastructure essential for sustainable transportation systems.

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